NUMBER 5 -- THEOLOGY THROUGH TIME AND SPACE

We will be dealing with several stories, all quite different yet with the unifying theme of COMMITMENT.

be watching as we do these for how that is worked out as well as for other themes that crop up.

~~the format for this class will be to give a synopsis of the story both for those who have not read them and as a review for those who have~~.

I will also give more background where it is appropriate, for example, on the Emmigrant we will need some background on the levitical priesthood.

I will ask for 2 of the stories to be read rather than give a synopsis, since they are short.

"The Emmigrant" ~~and one not on your syllabus called~~ and "Proselytes" by Gregory Benford.

We will discuss them as to theme, meanings, etc. ~~and I make no pretense of having the definitive answer but will referee the discussion as best I can.~~

2nd hour we will see clips from 2 movies that deal with commitment and fanataciesm. TT and Handmaids tale

ORDER;

We will begin with "THE STAR' written by A. C. Clarke in 1956 and won him the Hugo award for best short story that year. This rasies the problem of religious doubt to a man who felt space travel could never affect his faith.

2. "THE EMMIGRANT, 3. WHEN JESUS COMES DOWN THE CHIMNEY ~~4. PROSELYTES,~~ 4. THEOLOGY OF WATER. 5. Prosylytes

"THE STAR"by Arthur C. Clarke

This story is told from the viewpoint of a Jesuit priest who is considering what he, as chief astrophysicist, and his team, most of whom are not religious, have found in the Phoenix Nebula.

They are 3000 light years from Earth and it is the year approximately 2500 CE or so. He and his team have discovered some truth which upsets him and which he fears will end the Jesuits thousand year history and maybe much more than that. The Jesuits were founded in the mid 16th century by Ignatius Loyola, to whom he addresses his concern. a SUPERNOVA occurs about 3 times every thousand years or so. This is an explosion which is much more than a normal nova and creates a brilliant light in the sky. The chineese saw one in 1054, tho they were unaware of what it was, and another was seen in 1572.

The mission of this ship was to visit the remnants of one such supernova and learn its cause. They chose the phoenix nebulae but had no idea of the age until they arrived there. This one had occured about 6000 years earlier.

In exploring, they didn't expect to find any planets, but they discovered one at an emmense distance, the Pluto of this vanished solar system--to far from that sun to have known life.

They landed there and found a vault that was obviously the work of intelligent beings. The vault was built by a civilization that knew it was about to die. They had years to prepare and collect and bury all the fruit of their genius hoping someday someone would find it and they would not be utterly forgotten.

They could travel in their own solar system but could not yet cross the intersteller gulfs of space.

They were humanoid, as shown by their sculpture, and the thousands of visual records and machines for projecting them. From this the explorers got a picture of an advanced civilization as well as enough data to learn their written language.

The Jesuit's collegues question why, if God is merciful, was this civilization destroyed in the full flower of its achievement?

He believes his colleguws will answer that the universe has no purpose or plan and since a hundred suns explode in our galaxy every year, some race is dying at any moment. It makes no difference if that race is good or evil for there is no God and thus no divine justice.

However, he realized God doesn't have to justify his actions but can create or destroy when He chooses. It is arrogance to argue with God's decision or to say otherwise.

He says he could accept that, but because of the calculations before him, his faith falters. For he realizes that this particular supernova was the Star of Bethlehem and he must question why God would destroy a civilization to produce that symbol.

* * * * *

NOTES ON "THE STAR" - We must realize that A. C. Clarke is basically negative as far as Christianity, or other western religions, are concerned. In fact, he lives in Sri Lanka and seems to lean toward the THERVADA SCHOOL of Buddhism which says there is no one to help one acheive enlightenment, neither men nor gods.

For the Buddhist, enlightenment means to be free of desire by following the eightfold nobel path. Only then will one cease to suffer.

I say this only to show his viewpoint but this should not detract from the story or the idea involved which calls into question the ethics of God.

How do you react to this idea?

I feel there are some underlying assumptions here based on the Eastern idea of Karma, which would say we earn in previous lives what we get. However it also assumes reincarnation in the East. I would like you're input about that.

THE EMIGRANT - BACKGROUND

Priests in the Hebrew scriptures are representatives of the whole nation of Israel,

The whole people were to be holy to God but since that was an impossibility, the tribe of LEVI was set aside to be holy (Lev 11:44ff, Num 15:40)

THREE FOLD HEIRARCHY

All Levites are not equal:

The LEVITES were the lowest level, set apart for service in the sancturay, they represent the people of Israel as substitutes for the firstborn sons, who belong by right to God.(Num 3:12-13, 41,45; Exod 13:2, 12 -13; Lev 27:26, etc.)

SONS OF AARON are above these and specifically consecrated for the specific office of priest. They alone represent the nation in the sacrificaial ministrations of the alter.

THE HIGH PRIEST (Exod 28:29) bears the names of all 12 tribes on his breastplate. He alone can enter the Holy of Holies, and that only once a year on the DAY OF ATTONEMENT to offer sacrifice for himself and all the people (Lev 16:1-25)

In regard to our story THE EMIGRANT the priest is isolated from death, even of his relatives. (Lev 21:1-15)

The Israelite obsession with death was that death imparted a most severe impurity (Num 19:11-19). And if one was a priest, it was very much forbidden.

This may have been a violent rejection of the Egyptian religion that the dead and his buriel place were equal to the temple in their sanctity.

THE EMIGRANT, 1980

Joel Rosenberg

SYOPSIS OF STORY:

Yisro el Cohen, age about 30, a member of the hasidim, one of th present day Jewish mystical groups and also anORTHODOX JEW.

He is of the tribe of Levi and beard the name COHEN which makes him a member of the choenim, priest of ancient Judea.

The cohenim are still the first ones called to go to the alter and read from the Torah each Sabbath.

He is a butcher by trade, a SHAYCHET who knows how to butcher the meat so it is kosher.

He is the 3rd generation after the holocaust, living in the year 2000, on the Christian calander, oror 5750 on the Jewish.

The context of the story is in giving explanation to his brother in law, Morin, who is a catholic, why he must leave.

As he says, for the catholic, the final authority is the pope but for the Jew it is the law, or the TORAH.

He relates the history of this century beginning with Hitler's election in 1933 when 10 million or so Jews lived in Europe.

Yisro el has never been to any of the camps because it is forbiddend for a cohen to enter a graveyard.

However, the ashes from all those burned in the crematori have been spread by the wind all over the world. Every time people breath they are breathing in traces of all those killed and burned.

He says he must leave for since the world itself has become a graveyard he, as cohanim, must nt stay on Earth but will go to a colony on the moon as a butcher. Is he right? or is he overreacting? Is this also a commentary on our ecological problems?

The Emigrant

Joel Rosenberg

Joel Rosenberg was born in Manitoba and lives now in Connecticut, but part of his childhood was spent in a place called Northwood, North Dakota, where, he says, "there were two kinds of people—at least according to the strict Lutherans who were the vast majority: 'Papists,' who were Catholics, and Protestants, everybody else, including, I suppose, Jews, Baptists, Moslems, and people who sacrifice bulls to Jupiter on their living room rugs."

Rosenberg began writing in 1980. "The Emigrant" is his second sale. It's the shortest story in this book, but it's among the most powerful.

After supper, and his announcement, and his older sister's almost hysterical recriminations and protestations over his desertion, Yisroel Cohen stepped out the back door, stood on the damp evening grass, and tilted his face toward the heavens.

He was not an old man, no more than thirty years of age, but he had the beginnings of gray in his hair and beard, and the hints of lines to come in his face that seem to characterize Hassidim, even when young.

Idly, Cohen twirled an earlock around his index finger as he squinted at the sky. Above his head, looking like a fourth star adorning Orion's belt, he saw a glint of light that might be the Colony—no, it was moving too fast; most probably a satellite in low orbit. He tilted his head back, accidentally dislodging the black wool skullcap from its top.

As he stooped to pick it up, he heard the back door whisper open, and the heavy footsteps of his brother-in-law thud on the steps.

"Sarah sent me out to talk to you," Alfred Morin said. As with most non-Jews—or most Jews, for that matter—Morin was uncomfortable around Hassidim, Cohen thought.

Morin shifted his feet in the grass, uncomfortably.

Cohen straightened easily, saying nothing, his thin face holding only a trace of expression, perhaps weariness, perhaps discomfiture. He had not been looking forward to this weekend.

"What are you doing out here?" Morin asked.

"Seeing if, perhaps, I could spot the Colony," Cohen said. "But," he sighed, "it's too early, I guess."

He turned to Morin. "My bags are packed. After Shabas I will be going to the airport."

"And the house? What are you going to do with the house?"

Cohen shook his head and shrugged. "It really doesn't matter. You and Sarah may do with it what you please. The papers are at my lawyers'." He walked over to the oak tree in the middle of the yard, turned, and leaned his back against its bulk. Morin followed, and faced him.

"And the butcher shop?"

"Avram is taking it over. . . . He's my uncle's son. I don't think you've met him."

"No," Morin shot back, "that part of the family still doesn't accept me." He kicked at the base of the tree with his square-toed shoes. "I thought you were different."

Cohen chuckled, a strange sound in his throat, something halfway between a laugh and a sob. "No, it isn't you. Now, mind you, for my parents, it might have been. They would have said *kaddish*—the prayer for the dead—over Sarah, and never allowed either of you in their house."

Cohen laid his slim hand on Morin's muscular arm, and led him over to a pair of wicker lawn chairs. They sat down.

"But me? No. I am of a different generation from my parents, as they were of a different generation from theirs, who died—many of them and may they rest in peace—at the hands of Hitler."

Cohen sat, silently. Morin reached into his jacket pocket for a pack of cigarettes, offered one to Cohen, who declined with a motion of a finger, and lit up.

The tip of the cigarette glowed in the darkness.

"No," Cohen went on, as though he hadn't paused, "I am not responsible for how my sister lives her life—although she seems, still, to feel responsible for how I live mine."

"She did raise you, after the accident," Morin said, brushing his limp blond hair into place with his free right hand. "Old habits die hard."

"This you're telling to a Jew? After all, because I am named Cohen, after the *cohenim,* the priests of ancient Judea, congregations are still, after two thousand years, in the *habit,*" he bit the word off, "of letting me share, with other Cohenim, the privilege of first *aliyah,* of being first to go up to the altar for the reading of the Torah.

"Because I am a Cohen,"—he pronounced it Ko-hain—"I am not permitted to marry a widow, or be in a graveyard except to bury a blood relative, or another score of piddling things."

"But," Morin said, gesturing with blunt, nail-bitten fingers, "you seem, except for the way you dress, so, so *modern*, in so many ways."

"Because I call myself a butcher, instead of a *shaychet*, like my father and grandfather? No, I am not a modern man; I am an orthodox Jew, and sometimes I feel very out of step with this year 5750, the two thousandth year of the Christian era. I still," he said, finger-flicking the fringes that depended from underneath his shirt, "wear the *tzitzit*, I pray three times a day, and I study with Reb Gershon—who, by the way, I suspect my sister blames for all of this, for my leaving."

Morin took his cigarette between thumb and forefinger, flicking it off in a shower of sparks into the darkness. "So why *you?*" he asked. "Why not everyone named Cohen?"

"Because I'm a Jew, and you're thinking like a Catholic." He raised a peremptory palm. "No, I don't mean that as a put-down. You are a good man, Alfred, and you treat my sister well.

"But . . . were I a Catholic, and had a . . . religious problem, I would ask a priest. If he did not know, he would ask a bishop, who, still puzzled, would ask a cardinal, I suppose, all the way up to the pope, who is the sole arbiter of what is right and wrong. For Catholics.

"But for a Jew? No, no final source, except the Law, and the Almighty. And the conscience.

"And," he sighed, "the possible. To wish for, or attempt the impossible," he said, warming to the subject, "is a sin against God, who reserves the impossible for Himself.

"The day my parents died in the crash," he mused, "and I wished it wouldn't have happened—I sinned." Cohen shook his head, wondering. "That I really don't understand."

"I don't either," Morin said, lighting another cigarette and shifting uncomfortably in the hard chair. "And I still don't really understand why . . ."

"Just so," Cohen said. "In there," he indicated the house with a shrug of his thin shoulders, "I stated the bare facts of my leaving—mainly to get it out of the way, partly for my sister's benefit, because she should understand. For you I will . . . trot out some history.

"Please understand, I am a butcher, not a historian, but this bit of history every Jew knows." He leaned forward, resting his elbows on thin, bony knees, and rested his forehead in his cupped hands.

"In 1933, the year that the Germans elected Hitler, there were ten million Jews in Europe.

"Between 1939 and 1945, the Nazis murdered millions of people in their death camps. Six of those millions were Jews." Cohen spoke slowly, carefully, as though each word were being written down.

"Some were forced to dig huge pits, lined up on the edge, and shot. Films—made by the Nazis themselves—survived the attempt to cover up in 1944 and '45 when the records were being burnt, when plans were made to destroy and cover the camps themselves.

"Most of the people, though, died in gas chambers, where they were told, half-heartedly by the sniggering guards, that they were going for 'decontamination.' They were crammed inside, the doors locked and barred, and crystals of Zyklon-B were dropped in through hatches in the ceilings.

"Their death was not a pleasant one. I have never been to Auschwitz—I am a Cohen, and may not enter a graveyard—but I have seen pictures of the gas chambers. The walls, the floors, and even the ceilings are marked with deep scratches from fingernails; people trying, in their last agony, to claw their way through the concrete of the room, perhaps to leave some mark of their passing.

"After the bodies were looted, they were burned." Cohen's voice was level, as always. Morin's eyes stung as he lit another cigarette. "Some in crematoria, some on open steel frames where the air could circulate more easily, as the corpses were splashed with gasoline, and set on fire.

"Some of the ashes were buried in deep pits, some scattered in the winds.

"And the winds blow across the face of the world. Every time we breathe, we breathe in traces of these people, murdered and burned."

Cohen sat silent again. Morin was almost sure that he had finished, and had opened his own mouth to speak when Cohen continued.

"And now, a quarter of a million miles from here, men build the Colony from aluminum and steel taken from the moon.

"I have applied for a job there—they will need butchers—and I have been accepted, and must go.

"For I am a Cohen, descended of priests, and to be in a graveyard is forbidden me."

"WHEN JESUS COMES DOWN THE CHIMNEY" by Ian Watson

This next story brings up the issue of what can happen to truth given enough time and immagination. The crucial thing here is given the time between the events and the recording, if any of these events, what can happen to the real story. This is exagerated, of course, but the fact is that the gospels were set down at least 2 generations after Jesus life and death and I believe he wants to make the point that there could be fallicies in the story, tho not as blatent as this one.

Father tells son the story of when Santa was born in a stabel and the 3 magicians hiked a thousand miles to be there when he was born. The follow a bright comet in the sky and bring a magic sack as a gift.

Santa's mother, to prevent jealousy by the neighbors and to keep the greedy emporer from getting it, hides it.

When Santa became a man, she gave it to him. He decided to shower presents from the sack on the country folk, tho not ever take anything out of it for himself.

So he began tramping around giving people what they wanted or needed. Even so, others asked for things (a fur lined coat, a pair of black boots) and make him take them. Food was also pushed on him so he became very stout.

The romans arrested him as he was de-stablizing a marginal economy. He was crucified with the sack over his head. When dead he was stuffed into his sack and buried by his friends in a tomb.

When robbers broke into the tomb to steal the sack, it was

[illegible] th

empty, as if it had swallowed him. The robbers were awed and began to spread the word @ Santa all over the world, carrying snippets of his sack with them as proof.

Most of the sack ended up in Torino (Turin?) where it remains to this day. Thus when everyone hears of Santa and loves him, the sack will again distribute free gifts.

On Easter, gifts are distributed wrapped in sack cloth in mimory of Santa.

STORY OF JESUS

Tonight, the father says, is the one important to Jesus.

He was the leader of those theives who broke into Santa's tomb. "Robbery," he says, "is the product of a society where there arn't enough gifts to go around- -or where there are to many gifts for to many people."

Jesus is the one who carried the sack to Roma where he went straight to the senate and began preaching: (quote on pg92).

"Your desires are the thesis. This sack is the antithesis. The synthesis is that you should empty yourselves of false goals, vain dreams, the products of a diseased society. Just you empty all of that false consciousness of yours into this sack! It will hold everything, and reduce everything that is contradictory. In its place you'll discover that gifts ought to be given according to one's needs, not one's desires--but society at present is based on legalized theft, on the alienation of persons from their soil, from their work, even from their own bodies and sexuality."

I believe this is Watson's thesis. How do you react to this statement? It has more to do with Buddhist teaching than with what Jesus taught. Do you agree?

Soon many plebians began to believe him and stepped into the sack and out as a sign of a change of heart.

The emporer took notice, going to the forum intending to spear Jesus, but on the way had a vision seeing a sack in the sky which swallowed the sun (probably a total eclipse). On arrival he stepped into the sack and the empire became a republic.

(This is, of course, Constantine in 312 CE)

Tonight Jesus will come down the chimney and take awaoy whatever you think is most precious to you. This is how other deserving children receive gifts at Easter.

Jesus will re distribute all our wealth. He brings Santa's sack empty down every chimney and fills it full from every house.

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I think there are several purposes in this story, which is one of the reasons I used it.

First, we have Watson showing what can happen to a story, any story, given enough time. This story takes place, presumably, in some far future when Santa has become more important than Jesus. The ultimate commercialism. Is this possible? or has it happened already? Is this a comment on our consummer society?

Also, given that the first stories of Jesus were not written down until between 70 and 90 CE, at least 2 generations after his death/resurrection, I think he is saying that we can't totally rely on what is considered the "gospels". Things get exagerated, used for polemical purposes, etc. With any oral tradition, as we all know from the game of "gossip", things can change over very little time, much less over a long period.

THESIS I believe is his statement given as preaching of Jesus. It has more to do with Eastern thought, especially Zen Buddhism, than with what Jesus taught.

What do you all think?

Can you see any other purposes for this story?

THEOLOGY OF WATER - SUMMARY © 1980 - Hebert Schenck

NASA sends a team to Titan, one of Saturn's moons, to look for life.

the team is made up of comd. Brunel, an astoronomer and pilot, Dr. Bateson, a chemist, Dr. Feeney a Jesuit priest and geophysisist, and Dr. Takoa a Japanese amer. exobiologist.

-found water that froze at -67.1 celsius (instead of 0)
-Earth water and Titan's water's curve matched exactly
-Titan ice floated higher than norman ice, thus lower density

Fr. Feeney predicted exactly how titan's water would behave then tells his baffled fellow team a story of a fellow Jesuit, Fr. Fullerton, whose doctoral disertation predicted the way water would behave on every kind of planet. His dissertation was nicknamed "the Theology of water."

he varied the water properties for our given Earth then tried the true water properties on planets at different locations, with different masses.

the tilt of the Earth makes all the difference as to what happens to the properties of water. If the Earth were exactly straight, there whould be to much stability, constant cloud cover, and ice caps down to latitude 40 degrees north and south.

if the polar axis is flat, as Uranus, there would be wild storms constantly.

conclusion: the properties of water on earth were geared to us humans, beings that operate at 98.6 deg. F.

- the chances of a water-optimized Earth occuring in our galaxy ~~were one in 10 sats the 7 or 9th~~ was so small that Earth was it.

-fullerton's conclusion was that we on Earth were all ther was anywhere. This depressed him so much he committed suicide.

Fr. Feeney quit the university, went to retreat in NM and then joined NASA.

chi- .

Even tho Schenck disclaims that Fullerton uses the dissertation to refute Copernicus, he in fact does just that by saying the creation is for us humans. When we arrive at a planet, it will change to suit humans so humanity is still the center of the universe.

His approach is basically gnostic for he says to be a saint is to KNOW. This is not the Christian definition of sainthood for most saints did not feel the knew.

This has the same problem that Gnosticism originally had. Ie, some people are capable of knowing and thus are saved while others were believed to not be capable of this knowledge and thus had no hope of salvation.

PNEAUMATICAS where those spiritual humans who when given the knowledge that leads to salvation would automatically accept it and be saved.

PSYCEKOS are those humans who were "soulish" (as opposed to spiritual) who might be saved but it would take a lot of work to learn the knowledge (gnosis) necessary for salvation. however, they at least had hope.

HYLIKOS were those who were carnal, so much so that they had no hope of salvation.

this scheme was developed by Valentinius in the 2nd cent and was rejected by the church in favor of belief in Christ not a particular knowledge that could be acquired.

Thus Hilbert Schenck uses the science fiction genre to promte his own religious ideas rather than dealing with science fiction.

however, Dr. Takoa sees another interpretation. IE, the Earth's water was specifically designed for it as kTitan's water was for it.

They run another test on Titan's water, what they had distilled earlier. They find it is now identical to Earth's water.

Feeny theologizes. God designs each world, tho on most nothing comes to live, but when humans arrive on Tital, they are the life and he has given Titan, its water and everything to them.

God is Johnny Appleseed preparing orchards across the universe and changing them when a differend seed comes by.

Feeny understands how one becomes a saint: you KNOW.

they checked the ice sheet again, pulling a sample.
-Earth water again
-something has changed the entire planet.
finally, hundreds, maybe thousands of other groups were making the same discovery and responding the same way this group did, with awe.

the entire universe was constantly pierced by bright, new asrrows of surprize, love and joy.

* * * * * * *

[illegible] develops [illegible] the [illegible] same.

This story is using a scinetific context and then suspending science in favor of "bright arrows of surprize, love and joy."

It handles science in an entirely different manner so the basic premise about science is dramatically different from most. Traditional Science Fiction

Comments!

"PROSELYTES" BY Gregory Benford, Late 80's

one of the finest hard Science fiction writers, he has won the nebula Award twice, as well as the British Science Fiction award and others. This story is a change of pace for him.

SYNOPSIS: Seven foot tall beings called Gacks have arrivaed and are bringing the "glorious word from the stars" and the "True knowledge of the Universe".

They arrived in astroid shaped ships using the simplest methods, small nuclear bombs exploded out the back. In return for allowing them to speak, they will share their technology.

Unfortunately, Earth is already ahead of them since they still use vacuum tubes, hand crank computers, etc.

They are analogous to the Jehovah's witnesses and Mormans, passing out literature and giving their spiel.

At one home, after listening to their spiel, Dad slams the door and comments that the only kinds of aliens who're damn fool enough to spend all their time flying to the stars are not explorers. not scientists. but fundamentalist.

Suddenly the sky is alive with darting ships blowing up the Gack Ships. They are sleek, shiny vessels, and junior says: "Thats what an alien starship oughta look like."

A banner unfurls from one of the ships reading: GREET THE CLEANSING BLADE OF THE ONE ETERNAL TRUTH"

Huh! says Junior, Dad's face goes white and he says: "we thought the Mormons were bad, whoever thought there might be Moslems?"

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This story is in the realm of what if: What if the first contact with an alien or more than one alien civilizaiton turns out to be not sceintists or explorers, but religious fanatics determined to convert humans to the "one true and right way?"

worse, what if their philosopy is "conver or die?"

Proselytes
Gregory Benford

It was the third time something had knocked on the door that evening. Slow, ponderous thuds. Dad answered it, even though he knew what would be standing there.

The Gack was seven feet tall and burly, as were all Gacks.

"Good evening," it said. "I bring you glorious word from the stars!"

It spoke slowly, the broad mouth seeming to shape each word as though the lips were mouthing an invisible marble. Then it blinked twice and said, "The true knowledge of the universe! Salutations of eternal life!"

Dad nodded sourly. "We heard."

"Are you certain? I am an emissary from a far star, sent to bring—"

"Yeah, there's been two others here tonight already."

"And you turned them away?" the Gack asked, startled.

Junior broke in, leaving his homework at the dining-room table. "Hey, there's been *hundreds* of you guys comin' by here. *For months.*"

The Gack blinked and abruptly made the sound that had given the aliens their name—a tight, barking sneeze. Something in earth's air irritated their large, red noses. "Apologies, dear ignorant natives, from a humble proselyte of the One Patriarch."

Dad said edgily. "Look, we already heard about your god and how he made the galaxy so you Gacks could spread his holy word and all, so—"

"Oh, let the poor thing finish its spiel, Howard," Mom said, wiping her hands on a towel as she came in from the kitchen.

"Hell, the Dodgers' game'll be on soon—"

"C'mon, Dad," Junior said. "You know that's the only way to get 'em to go away."

The Gack sniffed appreciatively at Junior and started its rehearsed lecture. "Wondrous news, O Benighted Natives! I have voyaged countless of your years to bring . . ."

The family tuned out the recital. As Dad stood in the doorway he could see dozens of Gack ships orbiting in the night sky. They were like small brown moons, asteroid-sized starships that had arrived in a flurry of fiery orange explosions. Each had a big flat plate at one end. They were slow, awkwardly shaped, clunky—like the Gacks themselves.

They had come from a distant yellow star and all they wanted was free reign to "speak to the unknowing," as their emissary had put it. In return they had offered their technology.

Dad had been enthusiastic about that, and so had every government on Earth. Dad's half-interest in the Electronic Wonderland store downtown had been paying very little these last few years. An infusion of alien technology, whole new racy product lines, could be a bonanza.

But the Gacks had nothing worth using. Their ships had spanned the stars using the simplest possible method. They dumped small nuclear bombs out the back and set them off. The ship then rode the blast wave, with the flat plate on one end smoothing out the push, like a giant shock absorber.

And inside the Gack asteroid ships were electronics that used vacuum tubes, hand-cranked computers, old-fashioned AM radio . . . nothing that humans hadn't invented already.

So there would be no wonder machines from the stars. The sad fact was that the aliens were dumb. They had labored centuries to make their starships, and then ridden them for millennia to reach other stars.

The Gack ended ponderously with, "Gather now into the outstretched loving grasp of the One True Vision!"

The Gack's polite, expectant gaze fell in turn on each of the family.

Mom said, "Well, that was *very* nice. You're certainly one of the best I've heard, wouldn't you agree, Howard?"

Dad hated it, how she always made *him* get rid of the Gacks. He began, "Look, we've been patient—"

"*They*'re the patient ones, Dad," Junior said. "Sittin' inside those rocks all those years, just so they could knock on doors and hand out literature." Junior laughed.

The Gack was still looking expectantly at them, waiting for them to convert to his One Galactic Faith. One of its four oddly shaped hands held forth crudely printed pamphlets.

"Now, now," Mom said. "We shouldn't make fun of another creature's beliefs. This poor thing is just doing what our Mormons and Jehovah's Witnesses do. You wouldn't laugh at them, would you?"

Dad could hold it no longer. The night air was cold and he was getting chilly, standing there. "No, thanks!" he said loudly, and slammed the door.

"Howard!" Mom cried.

"Hey, right on, Dad!" Junior clapped his hands.

"Just shut the door in its faith," Dad said, making a little smile.

"I still say we should always be polite," Mom persisted. "Who would've believed that when the aliens came, their only outstanding quality would be patience? The patience to travel to other stars. We could learn a thing or two from them," she added sternly.

Dad was already looking in *TV Guide*. "We should've guessed that even before the Gacks came. After all, who comes visiting in this neighborhood? Not that snooty astronomer two blocks over, right? No, we get hot-eyed guys in black suits, looking for converts. So it's no surprise that those are the only kinds of aliens who're damn fool enough to spend all their time flying to the stars, too. Not explorers. Not scientists. Fundamentalists!"

As if to punctuate his words, a hollow *thump* made the house creak. They all looked to the front door, but the sound wasn't a knock.

Another *boom* came down from the sky and rattled the windows.

When they went outside, the night sky was alive with darting ships and lurid orange explosions.

Junior cried, "The Gack ships! See, they're all blown up!"

"My, I hope they aren't hurt or anything," Mom said. "They're such *nice* creatures, truly."

Among the tumbling brown remnants of the Gack fleet darted sleek, shiny vessels. They dived like quicksilver barracuda, send-ing missiles that ripped open the fat bellies of the last few asteroid ships.

Dad felt a pang. "They were kinda pleasant," he said grudgingly. "Not my type of person, maybe, and their technology was a laugh, but still—"

"Look!" Junior cried.

A sleek ship skimmed across the sky. A bone-rattling *boom* crashed down from it.

"Now *that's* what an alien starship oughta look like," Junior said. "Lookit those wings! The blue exhaust—"

Behind the swift craft huge letters of gauzy blue unfurled across the upper atmosphere. The phosphorescent words loomed with hard, clear purpose:

GREET THE CLEANSING BLADE OF THE ONE ETERNAL TRUTH!

"Huh?" Junior frowned.

Dad's face went white.

"We thought the Mormons were bad," he said grimly. "Whoever thought there might be Moslems?"

What, if any is the overarching theme of these stories?

COMMITMENT IN OUR STORIES

"The Star", of course, has a very committed Jesuit, who thinks there is nothing that can hurt his faith in God. However, upon discovering that the super nova which became the star of Bethlahem destroyed a whole civilization, he is not only shaken, but believes that all those of his order, and for that matter, most Christians, will in some degree or another, lose their faith.

The opposite number to this story is "Theology of Water" in which the jesuit, who is losing his commitment and faith, finds it reinstated upon discovering that God has planted the potential for human life throughout the universe. Moving from depression that the Earth was all there was anywhere for human habitation to the fact of God creating a habitat for humans everywhere caused a veverseal of his viewpoint and his depression.

"The Emmigrant", if anything, is showing over commitment to his calling of cohanim or priest. I doubt that many cohanim whould go this far, but as we travel and perhaps colonize, the moon and other planets, who knows who the thinking may develop within Judaism.

As for "When Jesus Comes down the Chimmney" I think it shows how one can be committed to some crazy view of reality, no matter how far out it may be. Carl Sagen in his [redacted] book, <u>The Demon Haunted World</u> believes that many people today are involved in, or committed to, ideas which have no scientific or even rational basis. We have traded UFOs and Aliens from outer space for monsters that dwell in the world or the sea. However, the attitude is similar, in that we project our fears onto the unknown. In that most humans have not changed.

"Proselytes" I feel, is fairly self exclamatory. The Gacks and the psudo-muslims are among the super-commited, and, as the story tries to show, it is this very committemnt that motivated them out into space and may be these who we make first contact with in the future, rather than the scientist or explorer.

Pi and Handmaid's tale

Now I want to deal with some clips from two movies which also deal with the theme of commitment which can lead to obsession, madness and even to fanaticism.

PI - won sundance director's prize in 1988. on a budget of only 60,000 dollars.

The first is the movie PI, named for the Greek letter which equals the ratio between the circumfrance of a circle to its diameter. But the movie is less about mathamatics as it is a study in obsession, a visual representation of madness. — go to next page @ Pi →

It's protaganist, Max Cohen, believes that mathamatics is the true language of nature and that every action in the world, no matter how random, is intricately ordered and patterened. Given this thesis he decides to use this knowledge to try to predict the stock market. (show scene 3, "state my assumptions)

The film goes beyond this initial premise and takes it to the plane of faith. Max is an agnostic Jew but he draws the attention of Lenny Meyer, a Jew from a sect involved in Kaballah, the Jewish mystical tradition. Lenny explains that the hebraic language in numerical and replete with symbolism, with the key to the meaning of life. (watch exchange scenes 9 and 10 Subway notes and Kaballah)

Max's computer occupies most of his apparment. When it crashes it leaves behind a printout of random numbers which he throws away. The question is: are these numbers the secret 216 digit number that the Kaballah sect is looking for?

The meaning of the number is clearly explained in this next clip (show 31 - 35 Rabbi Cohen to the end)

This film is interesting because it starts out atheistic and orderly and shifts its tone; that maybe God exists and He could be explained thru science; until the stunning conclusion that perhaps God exists and is unexplainable.

* * * * * * * *

Pi

There was a brief period in my life when I was obsessed with the universal constant Pi just as much as Max Cohen (Sean Gullette) was obsessed with finding the number that was the answer to the universe in *Pi*. That period was during my last undergraduate semester, where we devoted our efforts to estimating Pi to as many digits as possible for our Numerical Analysis class, which culminated in a conceptual painting about the irrational.

Pi isn't about the mathematical constant 3.1415926..., representing, among many other things, the ratio of the circumference of a circle to its diametre. The movie is about the deranged and beautiful quest of one person in search of the truth, the answer to the universe. The plot is a common one in science fiction: a phenotypic "aberration" in the brain causes the protagonist to develop special abilities that makes him sought after and feared. In this particular cases, Max acquires a deep grasp of number theory. With his assumptions, that mathematics is the universal language, that number theory can represent everything in nature, and that there is a pattern in everything that occurs in this universe, he sets about trying to find it in the stock market.

After him are people who are interested in his stock market analyses for monetary purposes, and more strangely, a group of Rabbis who are convinced the same pattern of numbers is the key to their salvation. However, Max is the only one who can understand the semantics of the 216 digit number that is key to the universal lock, a plot device that I thought was truly brilliant. In the end, Max succeeds on his quest, but what he understands is never revealed to us. What he sees however is catalyst enough for him to inflict a lobotomy upon himself.

Today, many scientists (~~including myself~~) are on the same quest that Max is, and most don't need to be as obsessed to find what they're looking for. As demonstrated several times over the course of humanity's existence, there is indeed an explanation for a lot of things we see in nature. That is, at least a conceptual level, there do exist patterns that can be written out as mathematical equations. This is particularly true given the discoveries this centuries involving relativity, quantum mechanics, and the subsequent biological and computing revolutions. But the search for the Grand Unified Theory continues.

Even though the movie isn't directly about Archimedes' constant, *Pi* is indeed an excellent solution to Max's problem. That constant is ubiquitous in our world today, popping up every so often, in a seemingly independent manner. Besides its strong presence in geometry, Pi appears in various equations throughout mathematics (especially certain infinite series) and even in places where you wouldn't necessarily expect it to at first intuition. For example, the Buffon needle problem: what is the probability that a needle of length 1, thrown at random on a plane divided by parallel lines 1 unit apart, will land in such a manner that it crosses a line (2/Pi)? What is the probability that two integers chosen at random have a common factor exceeding one (6/Pi^2)?

Unlike many independent films, *Pi* actually has a fairly coherent, albeit obscure, plot. The cinematography is excellent as well. Filmed completely in black and white, with obscure and surreal settings, the movie tries very hard to brings the audience to the edge of the abyss that is Max's mind. The acting by go Bach
Sean Gullette is highly convincing. The electronic music (courtesy of Clint Mansell of Pop Will Eat Itself) is top-notch. The movie is definitely worth the approximately Pi * e dollars I paid for it and I highly recommend checking it out at a local independent theatre near you.

Movie ram-blings || Ram Samudrala || *me@ram.org*

<u>Handmaid's tale</u> (1990, Faye Dunaway, Robt Duvall, Aidan Quinn, Natasha Richardson)

Based on the book of the same name by Margaret Attwood, this looks in the not to distant future in a "what if" situation. That being that the US becomes a theocracy. This is a sciecne fiction look at what might happen if the right wi ng were actually to come into power. This is a case of commitment turning into fanaticism.

The scary part of it is that there is a movement, the ~~Restoration~~ Reconstructionist Movement it is called, in this country that transcends denominations but are agreed that the US should be governed by the Bible as interpreted literally. Many prominent ministers, including Rev. Criswell, the minister of the largest southern Baptist chuch in the country found in Dallas, think that women should not have the rights they do, that stoning should be the punishment for numerous offences following the Levitical laws of the Hebrew Scriptures. Bill Moyers was concerned enough about this movement he produced a two hour special on it a few years ago. today, Dominion Theology has similar motives - Raphael Cruz & son Ted, et al are part of this movement.

In the opening scenes, we find Kate and her family trying to escape over the border into Canada. (Show opening scenes I and 2 "the lucky ones")

Here are a couple of scenes showing the brainwashing the women go thru followed by the ceremony which handmaids must endure.

The last scene we will look at is the extreme of fanatical power. When anyone disobeys what this theocracy has deemed as law. (show scene 12, executions)

This movie relates mostly to the short story "Proselytes" and highlights the theological difference between Commitment and outright fanaticism. It can be a thin line

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Ronald Farmer: "Theocracy" - Kindle, 2.99

PROSELYTES

GREGORY BENFORD

It's difficult to avoid hyperbole when discussing Gregory Benford. He is regarded by most as one of the finest (if not simply *the* finest) hard science fiction writers working today, a writer who has introduced higher literary values to the subgenre than any before him. He has won the Nebula Award twice, along with the British Science Fiction Award, the John W. Campbell Memorial Award, and the Australian Ditmar Award. His widely-popular and highly-acclaimed novels include the Spectra titles *In the Ocean of Night, Across the Sea of Suns, Great Sky River,* and *Heart of the Comet* (which he wrote with David Brin) along with *Timescape, Artifact,* and others which unfortunately do not appear under the Spectra imprint. He will have a new novel out in early 1989 which, as we write this, is still untitled. Greg lives in Laguna Beach with his wife and two children and he is a professor

The mystery of Christmas is deepened far out in time and space.

The Star

Arthur C. Clarke

It is three thousand light-years to the Vatican. Once, I believed that space could have no power over faith, just as I believed that the heavens declared the glory of God's handiwork. Now I have seen that handiwork, and my faith is sorely troubled. I stare at the crucifix that hangs on the cabin wall above the Mark VI Computer, and for the first time in my life I wonder if it is no more than an empty symbol.

I have told no one yet, but the truth cannot be concealed. The facts are there for all to read, recorded on the countless miles of magnetic tape and the thousands of photographs we are carrying back to Earth. Other scientists can interpret them as easily as I can, and I am not one who would condone that tampering with the truth which often gave my order a bad name in the olden days.

The crew are already sufficiently depressed: I wonder how they will take this ultimate irony. Few of them have any religious faith, yet they will not relish using this final weapon in their campaign against me—that private, good-natured, but fundamentally serious, war which lasted all the way from

Earth. It amused them to have a Jesuit as chief astrophysicist: Dr. Chandler, for instance, could never get over it (why are medical men such notorious atheists?). Sometimes he would meet me on the observation deck, where the lights are always low so that the stars shine with undiminished glory. He would come up to me in the gloom and stand staring out of the great oval port, while the heavens crawled slowly around us as the ship turned end over end with the residual spin we had never bothered to correct.

"Well, Father," he would say at last, "it goes on forever and forever, and perhaps *Something* made it. But how you can believe that Something has a special interest in us and our miserable little world—that just beats me." Then the argument would start, while the stars and nebulae would swing around us in silent, endless arcs beyond the flawlessly clear plastic of the observation port.

It was, I think, the apparent incongruity of my position that caused most amusement to the crew. In vain I would point to my three papers in the *Astrophysical Journal*, my five in the *Monthly Notices of the Royal Astronomical Society*. I would remind them that my order has long been famous for its scientific works. We may be few now, but ever since the eighteenth century we have made contributions to astronomy and geophysics out of all proportion to our numbers. Will my report on the Phoenix Nebula end our thousand years of history? It will end, I fear, much more than that.

I do not know who gave the nebula its name, which seems to me a very bad one. If it contains a prophecy, it is one that cannot be verified for several billion years. Even the word nebula is misleading: this is a far smaller object than those stupendous clouds of mist—the stuff of unborn stars—that are scattered throughout the length of the Milky Way. On the cosmic scale, indeed, the Phoenix Nebula is a tiny thing—a tenuous shell of gas surrounding a single star.

Or what is left of a star . . .

The Rubens engraving of Loyola seems to mock me as it hangs there above the spectrophotometer tracings. What would *you*, Father, have made of this knowledge that has come into my keeping, so far from the little world that was all the universe you knew? Would your faith have risen to the challenge, as mine has failed to do?

You gaze into the distance, Father, but I have traveled a distance beyond any that you could have imagined when you founded our order a thousand years ago. No other survey ship has been so far from Earth: we are at the very frontiers of the explored universe. We set out to reach the Phoenix Nebula, we succeeded, and we are homeward bound with our burden of knowledge. I wish I could lift that burden from my shoulders, but I call to you in vain across the centuries and the light-years that lie between us.

On the book you are holding the words are plain to read. AD MAIOREM DEI GLORIAM, the message runs, but it is a message I can no longer believe. Would you still believe it, if you could see what we have found?

We knew, of course, what the Phoenix Nebula was. Every year, in our galaxy alone, more than a hundred stars explode, blazing for a few hours or days with thousands of times their normal brilliance before they sink back into death and obscurity. Such are the ordinary novae—the commonplace disasters of the universe. I have recorded the spectrograms and light curves of dozens since I started working at the Lunar Observatory.

But three or four times in every thousand years occurs something beside which even a nova pales into total insignificance.

When a star becomes a *supernova*, it may for a little while outshine all the massed suns of the galaxy. The Chinese astronomers watched this happen in A.D 1054, not knowing what it was they saw. Five centuries later, in 1572, a supernova blazed in Cassiopeia so brilliantly that it was visible in the daylight sky. There have been three more in the thousand years that have passed since then.

Our mission was to visit the remnants of such a catastrophe, to reconstruct the events that led up to it, and, if possible, to learn its cause. We came slowly in through the concentric shells of gas that had been blasted out six thousand years before, yet were expanding still. They were immensely hot, radiating even now with a fierce violet light, but were far too tenuous to do us any damage. When the star had exploded, its outer layers had been driven upward with such speed that they had escaped completely from its gravitational field. Now they formed a hollow shell large enough to engulf a thousand solar systems, and at its center burned the tiny, fantastic object which the star had now become—a White Dwarf, smaller than the Earth, yet weighing a million times as much.

The glowing gas shells were all around us, banishing the normal night of interstellar space. We were flying into the center of a cosmic bomb that had detonated millennia ago and whose incandescent fragments were still hurtling apart. The immense scale of the explosion, and the fact that the debris already covered a volume of space many billions of miles across, robbed the scene of any visible movement. It would take decades before the unaided eye could detect any motion in these tortured wisps and eddies of gas, yet the sense of turbulent expansion was overwhelming.

We had checked our primary drive hours before, and were drifting slowly toward the fierce little star ahead. Once it had been a sun like our own, but it had squandered in a few hours the energy that should have kept it shining for a million years. Now it was a shrunken miser, hoarding its resources as if trying to make amends for its prodigal youth.

No one seriously expected to find planets. If there had been any before the explosion, they would have been boiled into puffs of vapor, and their substance lost in the greater wreckage of the star itself. But we made the automatic search, as we always do when approaching an unknown sun, and presently we found a single small world circling the star at an immense distance. It must have been the Pluto of this vanished solar system, orbiting on the frontiers of the night. Too far from the central sun ever to have known life, its remoteness had saved it from the fate of all its lost companions.

The passing fires had seared its rocks and burned away the mantle of frozen gas that must have covered it in the days before the disaster. We landed, and we found the Vault.

Its builders had made sure that we should. The monolithic marker that stood above the entrance was now a fused stump, but even the first long-range photographs told us that here was the work of intelligence. A little later we detected the continent-wide pattern of radio-activity that had been buried in the rock. Even if the pylon above the Vault had been destroyed, this would have remained, an immovable and all but eternal beacon calling to the stars. Our ship fell toward this gigantic bull's-eye like an arrow into its target.

The pylon must have been a mile high when it was built, but now it looked like a candle that had melted down into a puddle of wax. It took us a week to drill through the fused rock, since we did not have the proper tools for a task like this. We were astronomers, not archaeologists, but we could improvise. Our original purpose was forgotten: this lonely monument, reared with such labor at the greatest possible distance from the doomed sun, could have only one meaning. A civilization that knew it was about to die had made its last bid for immortality.

It will take us generations to examine all the treasures that were placed in the Vault. They had plenty of time to prepare, for their sun must have given its first warnings many years before the final detonation. Everything that they wished to preserve, all the fruit of their genius, they brought here to this distant world in the days before the end, hoping that some other race would find it and that they would not be utterly forgotten. Would we have done as well, or would we have been too lost in our own misery to give thought to a future we could never see or share?

If only they had had a little more time! They could travel freely enough between the planets of their own sun, but they had not yet learned to cross the interstellar gulfs, and the nearest solar system was a hundred light-years away. Yet even had they possessed the secret of the Transfinite Drive, no more than a few millions could have been saved. Perhaps it was better thus.

Even if they had not been so disturbingly human as their sculpture shows, we could not have helped admiring them and grieving for their fate. They left thousands of visual records and the machines for projecting them, together with elaborate pictorial instructions from which it will not be difficult to learn their written language. We have examined many of these records, and brought to life for the first time in six thousand years the warmth and beauty of a civilization that in many ways must have been superior to our own. Perhaps they only showed us the best, and one can hardly blame them. But their worlds were very lovely, and their cities were built with a grace that matches anything of man's. We have watched them at work and play, and listened to their musical speech sounding across the centuries. One scene is still before my eyes—a group of children on a beach of strange blue sand, playing in the waves as children play on Earth. Curious whiplike trees line the shore, and some very large animal is wading in the shadows yet attracting no attention at all.

And sinking into the sea, still warm and friendly and lifegiving, is the sun that will soon turn traitor and obliterate all this innocent happiness.

Perhaps if we had not been so far from home and so vulnerable to loneliness, we should not have been so deeply moved. Many of us had seen the ruins of ancient civilizations on other worlds, but they had never affected us so profoundly. This tragedy was unique. It is one thing for a race to fail and die, as nations and cultures have done on Earth. But to be destroyed so completely in the full flower of its achievement, leaving no survivors—how could that be reconciled with the mercy of God?

My colleagues have asked me that, and I have given what answers I can. Perhaps you could have done better, Father Loyola, but I have found nothing in the *Exercitia Spiritualia* that helps me here. They were not an evil people: I do not know what gods they worshiped, if indeed they worshiped any. But I have looked back at them across the centuries, and have watched while the loveliness they used their last strength to preserve was brought forth again into the light of their shrunken sun. They could have taught us much: why were they destroyed?

I know the answers that my colleagues will give when they get back to Earth. They will say that the universe has no purpose and no plan, that since a hundred suns explode every year in our galaxy, at this very moment some race is dying in the depths of space. Whether that race has done good or evil during its lifetime will make no difference in the end: there is no divine justice, for there is no God.

Yet, of course, what we have seen proves nothing of the sort. Anyone who argues thus is being swayed by emotion, not logic. God has no need to justify His actions to man. He who built the universe can destroy it when He chooses. It is arrogance—it is perilously near blasphemy—for us to say what He may or may not do.

This I could have accepted, hard though it is to look upon whole worlds and peoples thrown into the furnace. But there comes a point when even the deepest faith must falter, and now, as I look at the calculations lying before me, I know I have reached that point at last.

We could not tell, before we reached the nebula, how long ago the explosion took place. Now, from the astronomical evidence and the record in the rocks of that one surviving planet, I have been able to date it very exactly. I know in what year the light of this colossal conflagration reached our Earth. I know how brilliantly the supernova whose corpse now dwindles behind our speeding ship once shone in terrestrial skies. I know how it must have blazed low in the east before sunrise, like a beacon in that oriental dawn.

There can be no reasonable doubt: the ancient mystery is solved at last. Yet, oh God, there were so many stars you could have used. What was the need to give these people to the fire, that the symbol of their passing might shine above Bethlehem?

You better watch out . . . he's coming again!

When Jesus Comes Down the Chimney

Ian Watson

Now, Jamie, if you don't go to bed when your Daddy tells you to, Jesus won't come down the chimney!

Oh, so you can't even *imagine* sleeping yet?

Saints! Tell you the whole story of Jesus—and of Santa Claus too? Why, that would take till nine.

Well, maybe . . . (No, I am *not* spoiling the boy!)

You just snuggle up in your chair by the fire there, Jamie, and listen to me. And I'll be carrying you upstairs before you've heard the half of it!

We'd better start with Santa Claus.

We all know how Santa was born in a humble stable amongst the chickens and goats. Most of his countryfolk were poor, and Santa's parents were no exception. No shoes on their feet, no fine cakes in the larder. No larders, often! A lot of those people lived in tents, and it got pretty cold in the winter. Three magicians had hiked a thousand miles to be present when Santa was born. They followed a bright comet in the sky, and brought a magic sack as a gift. You could take whatever you wished for out of this sack. Santa's mother didn't want to stir up jealousy amongst her neighbors, so she hid the sack away. Anyway, her country was being occupied by the Roman army. If the Romans heard of the magic sack she feared they'd take it away for their wild, greedy emperor.

When Santa grew to manhood his mother gave him the magicians' gift and explained all about it. Santa decided then and there that he would like to shower presents on his countryfolk, though he swore that he would never pull anything out of the sack for himself.

So Santa tramped around the land with the sack over his shoulder, giving people whatever their hearts most desired, or what they needed most. He kept his vow about giving nothing to himself. Even so, one widow woman requested a fine red coat trimmed with angora wool and then insisted that Santa should wear it, not she. A leper whose feet were rotted and crippled asked for a pair of stout black boots, and forced these on Santa.

That wasn't all. Such a number of grateful people pressed bread and cheese on Santa, from out of their meagre stocks, not to mention fish and fruit and meat and milk and wine—which he couldn't decently refuse—that within a few years he grew positively stout!

Well, the Roman soldiers finally arrested him. All of those free gifts that poured from Santa's sack were destabilizing a marginal economy. They were weakening the currency. They were causing job refusal in the colonial labour market.

The Romans tied the magic sack over Santa's head. They marched him up to the top of a hill and nailed him to a wooden cross, then jabbed their spears through the sack a couple of times to blind him.

When they took Santa down dead at last they bundled his corpse into the sack, tied it tight, and set an official seal on it. They debated tossing him into the nearby river, but eventually their captain allowed Santa's friends to carry him away to a tomb.

That night the tomb was broken into by robbers who hoped to steal the magic sack . . . and they found the sack lying there empty. It was as if that burlap bag had digested Santa

Claus! As if it had spirited him away to the dimension where all free gifts came from.

The robbers were filled with wonder, and didn't want to steal ever again. Instead, they made a pact to spread the word about Santa all over the world and to carry the sack (or snippets from it) wherever they went, as proof. The sack was first carried to Roma, then later to Torino, where most of it remains to this very day.

In later years the descendants of those original robbers promised that one day when everybody in the world had heard of Santa and loved him, the sack would begin to distribute free gifts again. That's why, every Easter, we all receive presents wrapped in sack-cloth, in memory of Santa. Jesus? Oh, yes, I'm coming to him. Of course I am, Jamie! It's Jesus who's important tonight.

Jesus was the leader of those thieves who broke into Santa's tomb. (There's something symmetrical, don't you think, between gifts and robbery? Robbery is the product of a society where there aren't enough gifts to go round—or where there are too many gifts for too few people. What's that? Sym-met-ric-al. It means . . . oh, it doesn't really matter, Jamie darling. Honest!)

Jesus was the ex-thief who carried the sack to Roma where the hysterical greedy emperor lived, guarded by his soldiers with their spears.

When Jesus arrived in Roma he went straight to the Forum. That's a sort of meeting place, like a Senate, but for the common people.

Jesus stood up on a marble block and waved the empty sack and called out—with the help of a translator, from Aramaic into Latin, "Plebeians of Roma, I bring you gifts!" (A plebeian was someone unemployed, living on free bread and enjoying free entry into circuses.)

At first the plebeians who thronged the Forum stared at the sack as eagerly as if they were looking up a girl's skirt.

When they saw that the sack was empty, many of them hooted and jeered. Others lost their temper and chucked pebbles.

But Jesus cried out, "The gifts I bring you are dialectical!" (This was a term which Jesus borrowed from the Greek philosophers.) "Your desires are the thesis. This sack is the antithesis. The synthesis is that you should empty yourselves of false goals, vain dreams, the products of a diseased society. Just you empty all of that false consciousness of yours into this sack! It will hold everything, and reduce everything that is contradictory. In its place you'll discover that gifts ought to be given according to one's needs, not one's desires—but society at present is based on legalized theft, on the alienation of persons from their soil, from their work, even from their own bodies and sexuality."

With daily repetition, Jesus' message began to sink in. Soon a few of the plebeians believed him—and stepped into the sack and out again, as a symbol of their change of heart. Then many.

At last the emperor's curiosity was piqued; for the circus seats remained empty, and the elephants and the trained apes which rode them wept. Also there was growing unrest among his soldiers at the prospect of yet another colonial war.

The emperor in person led a party of trusted guards to the Forum, intending to spear this Jesus. On the way there the emperor . . . now, we must tell the truth: he was a hysteric but he also cunningly sensed his own political and economic infrastructure ebbing away . . . the emperor experienced a visionary fit. He saw a sack in the sky which swallowed the sun. (Actually, we believe this was a total eclipse.) When he reached the Forum he dismounted from his horse—and stepped into the sack. Soon the empire had totally changed . . . into a republic.

Ah, now you're nodding off.

Let's go quietly, mm? Up up up to bed.

Tonight, night of nights, Jesus will climb down the chimney and take away whatever you think is most precious to you. Will it be your rocking horse? Or your toy bear? Or just your tin whistle?

Tush. How else could other deserving little children receive fine gifts at Easter time?

Hush. He'll take something from us all. Not just you, you dobbin. Maybe I'll lose my spinning wheel tonight. Maybe it'll be my purple velvet dress.

Jesus'll redistribute all our wealth. That's why he's called "the good thief." He brings Santa's empty sack with him down all the chimneys in the whole wide world, and fills it full from every house.

Here we are now, darling. Tuck up tight, and shut those eyes. No peeping, or he mightn't come.

The Theology of Water

Hilbert Schenck

Hilbert Schenck won instant acclaim a couple of years ago with his first short stories, collected in Wave Rider *in 1980. His first novel,* At the Eye of the Ocean, *was published in 1981 and reinforced his reputation as a fine and thoughtful writer whose work is solidly grounded in both hard science and a deep sensitivity for the fragile humanity of his characters.*

In "The Theology of Water," Schenck presents an isolated group of scientists with a scientific question that may or may not have a scientific answer.

NASA had finally come to the Titan Mission as a desperate, final throw of the dice. The robot Mars landers, with their little automatic diggers and warmed and watered flower pots, had grown nothing. The probes to Venus, Jupiter and the asteroid belt all sent back discouraging messages: sterility and either violent surface conditions or utter barrenness. But then the big Saturn orbiter, an expensive wonder in itself, had turned its instruments on the moon, Titan, and a tiny hope was born. Titan had a thick atmosphere, mostly methane and other hydrocarbons but some oxygen, the surface pressure at almost seven-tenths Earth normal. Pasadena altered the orbiter's flight to let it free-fall around Saturn near Titan and several surprises appeared. There was ice, seas of it, in fact. Although the surface was at a horrendous mean temperature of minus one-hundred-and-twenty Celsius, the bolometer at the prime focus of the onboard reflecting telescope told of substantial warm spots—volcanic, radioactive, or both—where the temperature estimated beneath the ice sheet was actually above zero C. Most exciting, there were electrical storms that scorched the atmosphere of Titan and actually set high-altitude "sky fires," where the local oxygen concentration was high enough for ignition with methane, and these further warmed the satellite. The exobiologists of NASA, who for years had been trying to make life in their labs from methane and electrical discharges, were filled with hope, and though the public had mostly lost interest in the life-in-space hunt, NASA slowly began to assemble the Saturn-mission manned vehicle in orbit, panel by panel, appropriation by appropriation.

After almost fifty years, NASA had finally concluded that the number of consecutive push-ups and miles-run-per-day were irrelevant criteria for selecting deep space crews. The four persons on the Titan lander were thus the oldest and most expert astronauts ever to travel beyond the Moon. Mission Commander David Brunel, pilot and astronomer, had years of deep space activity, as did Dr. Gregory Bateson, a professor of chemistry at a distinguished New England university. Dr. Thomas Feeney, S.J., often referred to in the press as "God's Astronaut," had walked on the moon and had flown ten months in synchronous Earth orbit making geophysical measurements. And the tiny, fifty-year-old Japanese-American exobiologist, Dr. Anna Takoa, dean of her odd profession, was the ultimate hope for the lords of NASA in their search for other life. Veteran of half-a-dozen flights, this small woman, less than five feet tall and weighing ninety pounds, was the only scientist who had actually created a "pseudo-life" in the test tube, though she had never been able to get her chemicals to assemble duplicates of themselves. As a handsome bonus, Dr. Takoa required only sixty-three percent of the life support used by the other crew.

The gigantic Earth-Saturn transit vehicle took almost a year to make the trip, but once near Titan, the actual descent of the lander from its storage bay turned out to be simple. The cold, soupy atmosphere of Titan readily and safely absorbed the lander's potential energy and Brunel was able to drift his vessel across Titan's surface and pick his spot. They set down on a flat, volcanic plane, adjacent to a great lake or sea of ice that stretched to the horizon. And though they had all seen wonders not even imagined by most persons, they now stared, transfixed, out the lander's ports and up through the deep, blue-black sky of Titan to huge Saturn and its brilliant ring structure.

Father Feeney, his large, red face momentarily illuminated by the thin sunlight reflected from Saturn, stared in wonder. "Well, Anna," he said finally. "It's awfully cold, but the view alone ought to be enough to stir up some life out here."

The little woman peered up at the stocky priest and smiled gently. "What a nice idea, Tom," she said, "that life might emerge in response to beauty. If only it were true."

Suddenly a lightning-set sky fire went off in Titan's upper atmosphere where gravitational separation increased the molecular oxygen concentration, and the flat, rocky ground around them was fitfully bathed in an even redder light. They heard the thunder and then felt the ignition boom as the lander shuddered.

"I'm glad those things don't happen on the surface," said Commander Brunel half to himself. "Where do you think the O_2 comes from, Tom?"

"Out of the rocks, Dave," said Feeney at once. "I can't see how the ice could ever disassociate at these temperatures. There must be a continual fall and accretion of water and hydrocarbon compounds from those ignitions."

The Titan lander, its stubby wings jutting out on each side, sat on four eight-foot legs. The little craft had only two crew compartments, the upper one for sleeping, eating, and piloting, the lower one for lab work. The crew had known from the orbiter data that the ground-level atmospheric disturbances on Titan would be small. The high gas viscosity at these cold temperatures, coupled with the small contribution of heat from the distant sun, damped any weather activity. Their protective suits thus needed to resist only the bitter cold, but were flexible and non-armored, operating as they did at Titan's surface pressure.

First priority was the search for life. Dr. Takoa and Brunel went off in the electric, three-wheeled land-runner to search along the shores and out onto the ice sheet near them, while Tom Feeney and Bateson poked around the rocks and dirt of a nearby "warm spot" they had located while flying across Titan's surface. The NASA exobiologists, led by Anna Takoa, had developed an arsenal of marvelous, life-seeking instruments; devices that could detect the smallest bit of heat from the metabolic activity of microscopic, one-celled animals, protoplasm detectors that reported the weak radio waves emitted by all living cells undergoing certain kinds of biological activity, hydrocarbon probes that acoustically detected "large" or "complex" molecules such as might be associated with living matter.

Houston had designed the mission for a sixteen-day stay on Titan, to correspond with one complete orbit of the satellite around Saturn. Everyone had agreed before the start that if these four specialists could not find some kind of life on Titan in that time, it did not exist. And so, when they were not eating or resting, they went out on Titan's dark surface, peering, measuring, digging, and hoping for some tiny sign of warmth or irritability.

After several days of continuous search, their disappointments began to grow. Titan seemed as sterile as Earth's moon. There was ice with water under it. There was some oxygen and there were various hydrocarbons in the weathered "dirt." But it was a chill, flat, dark world, and finally Father Feeney and Professor Bateson decided to spend a day in the lander organizing and assimilating their data and specimens. While Tom Feeney tried various plotting methods to separate and locate Titan's two magnetic poles, Bateson worked in the chemical hood with some of the dirt and water from the surface.

"Hey, Father. Can you spare a minute?"

Tom Feeney, glum and disheartened, sighed and walked himself across the lab floor to the hood on his wheeled, aluminum stool. "Amoeba?" he suggested in a rumbling, hopeful voice.

Bateson shook his long, craggy head, but he was grinning. "No, but something weird anyway, Tom. I hacked out about a gallon of ice from the edge of the lake." He had both hands in the enclosed hood and he pointed with a slick, latex finger at the properties analysis box. "There's a sample melted from that ice in the magic box, Tom, and I'm getting a freezing point of minus 67.1 degrees Celsius."

Feeney frowned. "Quite a depression! Salty?" he suggested.

Bateson shook his head. "What I thought. So I distilled half a liter." He pointed to a compact apparatus stuffed in one corner of the long hood. "Not much residue, mostly dust. I'll look at it later. That material in the magic box now is distilled Titan water."

"It's not water, Greg, if it freezes at minus 67 C," said Father Feeney, frowning harder. "What is that stuff? It's absolutely clear?"

Bateson's grin turned sly. "If the audience will now direct its attention to this corner of the stage." His plastic-covered finger indicated a small electrolysis apparatus. "Notice that we have gas traps over the two electrodes, busily collecting gas from the electrolytic decomposition of this . . . ah . . . stuff. Note especially that one gas appears in just twice the volume of the other. Clue one. Now I take this glowing stick, ignited in the bunsen flame, and plunge it into the gas appearing in the lesser amount." He lifted the glass cover off one of the collection burettes and plunged the glowing stick into the colorless gas. It glowed whitely, then flamed up. "Clue two," said Bateson, peering sideways at Father Feeney's round mouth and eyes. He then took the flaming stick, lifted the cover off the other burette, and held the fire at the mouth. A blue flame and a loud pop immediately followed. "Clue three," said Professor Bateson, taking his hands out of the gloves.

But Tom Feeney's initial surprise had been immediately replaced by something else, a wariness coupled with an excitement. "Greg, what else is odd about that . . . water?" he asked in a tight voice.

"Let's see what we get for specific heat." Bateson put his hands back in the hood and fiddled with the controls of the properties box. In a moment the monitor screen beside the hood flashed some numbers. "About 20 percent higher," muttered Feeney thoughtfully. "Greg, can we get its density curve? Let's see if the extremum at 4 degrees C is still around anywhere."

The box heated and tested the sample as requested and soon a full data curve of liquid density against temperature appeared on the monitor. Tom Feeney blinked at the output. "Look at that, Greg! The extremum is much deeper and at least 20 degrees above the freezing point."

"Let me check that nutty thing with some real water," grunted the chemist, but when he did, the curve exactly matched the Earth water reference line.

"Why is the Titan curve stopped at 28 C?" asked Feeney.

Bateson peered at the monitor. "Because that's the boiling point, Tom. I didn't ask it for steam properties."

"Of course!" breathed Tom Feeney, and it suddenly seemed to Professor Bateson that Feeney was peering right out through the side of the lander.

At that moment they heard the *whiss* of the gas into the air lock. In a minute the hatch in the center of the circular lab compartment opened and a small, silver figure climbed up. Anna Takoa was both tired and discouraged. The finding of life, any life, on Titan would be the ultimate triumph in the career of any exobiologist. The hateful sterility of the place, cruel and harsh as it was, upset her. Titan wasn't much but it was now the only hope left in the system. And she, the greatest of her exobiologist tribe, would eventually have to pronounce it a lost hope to the world press. Her science simply did not exist, at least in this system.

Dr. Takoa was followed immediately by Mission Commander Brunel in a much larger suit. And even this extrovert seemed quiet and lost, an investor who had gambled years of effort in an enterprise now seen to be without gain or profit.

Still, they could both sense that something had changed in the lab. Tom Feeney had showed that same, slumped resignation when they had left on the electric vehicle for another life search, but now he was standing straighter and his large face was flushed and tense. Even Gregory Bateson's New England reserve seemed frayed and his hands were twisting together.

The priest stared at the arrivals as they hung up their suits and connected them for recharge. "Greg," he said, turning, "give them the same magic act you gave me. But start with the electrolysis."

Nodding, Bateson pushed his hands back into the gloves and explained where he had found the water, how he had distilled it, and what he was doing now. After the flame-up and the little blue pop of an explosion, Bateson turned to Commander Brunel. "What would you say that stuff was, Dave?"

Brunel smiled at him. "Hey, you'll have to ask harder ones than that to flunk *me* out. It's water, Professor, just like in Chem 101."

"Exactly," said Bateson drily. "Anna, what are your thoughts?"

The little Japanese woman crinkled her eyes at him. "It is obviously *not* water. Otherwise you wouldn't be going through all of this," she said with a small laugh.

"Now that's not fair!" said Bateson. "Anyway, I now place a sample of the water in the magic box and give you . . . freezing point, density of the liquid, and boiling point at one At."

They stared long and silently at the curve until Anna Takoa turned to look at Father Feeney. "Father Tom," she said lightly, "do you understand this?"

Feeney sat facing them. "I may. Some of it, anyhow. But I think we should look at whatever other properties the analysis box can give first. I have to tell you a story that I think may relate to this, but you'll believe it more after you see for yourself how this material acts."

Brunel peered shrewdly at the older man. "Is it water, Father?"

Feeney nodded. "Yes, I believe it is Titan water."

"What other . . . changes . . . in its properties do you expect?" asked Brunel, his cool eyes peering at Feeney.

The priest smiled. "Yes, you should test me that way, Dave. All right. I think the solid phase will not only float, but have an unusually low density."

Greg Bateson put his hands back in the hood and soon the monitor showed numbers and uncertainties. Brunel read off the ice density himself. "Point-six-two. I'll say it's lighter!" He whistled, then said more loudly to the chemist, "Greg, I know that super little properties box cost NASA a couple of million, but would you actually float a little Titan ice on a little Titan water for us to see?"

Bateson immediately filled a beaker with water and opened the door to a small refrigerator at the back of the hood. He removed a sliver of ice with tongs, chopped it roughly cubical with deft knife strokes and popped it into the beaker.

There was no doubt about it; the Titan ice floated far higher than normal ice. And as it floated, it rapidly melted, runnels of water appearing throughout the underwater part. This caused the cube to topple over, but as it turned upside down, it kept on rotating, over and over, faster and faster, growing steadily smaller and less cubelike. They stared, astonished.

Finally Gregory Bateson rubbed his fingers into the corners of his eyes and said to the exobiologist, "Care for a couple of those ice cubes in your before-bed nip of bourbon, Anna?"

But Tom Feeney was intently watching the ice turn and melt. "Greg, is there any substance that when it melts has an ordered, molecular momentum exchange, say from recoil effects, as the molecule leaves the solid phase?"

Bateson shook his head. "No way, Tom. No solid is ordered like that. And the distances are too small. . . .

But the priest was muttering, lost in thought. "We never even thought of that. A high-level ordering during solidification. But what use is it?"

"All right," said Brunel to Tom Feeney with a rueful grin. "I give up on this property. Tell us another, Tom."

Feeney pursed his lips and thought for a moment about Titan. "The vapor pressure curve should be steeper at the high end. Normalize the two curves, Greg, zero to 100, then superimpose them."

Indeed, the Titan water curve was much steeper near its boiling point than the curve for Earth water. Encouraged and excited, Father Feeney rattled off a few more predictions. "Thermal conductivity of the liquid is probably about the same, as is viscosity. Specific heat of the ice, the solid phase, will be higher. Latent heat of fusion also larger but boiling latent heat lower, way lower maybe. The ice probably has different absorptive characteristics to electromagnetic energy with infrared penetrating deeper."

Almost an hour later, after all these statements were shown to be correct, the three of them stared silent and baffled at the priest, whose face now twitched with excitement. "I think we'd better hear that story, Tom, before I go completely nuts!" said Brunel.

"All right," said Tom Feeney, settling his large bulk squarely on the small stool. "Understand, it was a painful time for me and I never thought. . . . Well, as you know, I went to Notre Dame grad school after I joined the Society of Jesus. The Jesuits have always been big on physical science, sort of 'keeping an eye on the enemy,' I suppose, and I was a star in the geophysics department. So I became a professor there after my doctorate, and wrote my books and did all that business. I didn't like academic life much. NASA has some stupid people and does some stupid things, but at least nobody is claiming they're some kind of unique humanistic seers while behaving like hypocritical phonies.

"I was on the doctoral committee of a student named Elias Fullerton. He was a priest, a Jesuit too, so he immediately asked me to be his major professor since his other committee members were two laymen and a stuffy old Dominican who was department chairman. Fullerton was tall and scrawny, had a huge Adam's apple, was nervous, jerky, what Freud would have seen as a classic neurasthenic.

"His dissertation topic was as odd as he was. He wanted to study, really optimize, the characteristics of water in relation to the climate of the Earth. Oh, many people had noticed how the stranger properties of water—for example its high heat capacity, density extremum, and floating of the solid phase on the liquid—related to climate and ocean behavior. But fifteen years ago it had just become possible to run computer simulations that could actually predict climatic changes with hypothetical changes in the properties of air or water. Before then we didn't have the knowledge or the equations. Too much was empirical. Also, the machines couldn't access the memory fast enough to do a complete Earth.

"The dissertation title was some typically academic jargon, 'Climatic parameter variations of real and simulated Earth-type planets as a function of the liquid and solid phase properties of water.' Something like that. Around the department, we called it 'The theology of water.'

"Well, the thing interested me. It was certainly a project with some scope to it, so Fullerton wrote the proposal and we all signed it and we were off. He did the study both ways. That is, he both varied the water properties for our given Earth, then tried the true water properties on planets at different locations, with different masses and so on. That's how I guessed out the trends for Titan's water. I tried to remember what was the best kind of water for small planets far from the sun."

"And the Earth's water and the Earth itself optimized together?" asked Anna Takoa.

The priest nodded. "It was striking. For example, if you simply assumed a continuously increasing liquid density with cooling, without our water's highly unique extremum at 4 degrees C, all kinds of drastic things happen. The cold water now lies stagnant in the deep oceans and the mean Earth temperature is over 20 Celsius degrees lower. Storms, wind and heavy cloud result. It's even worse if you make the ice sink instead of float. The oceans permanently freeze to within a couple of hundred feet of the surface. The rapid air temperature changes lead to massive storm systems. Fullerton's simulation predicted winds averaging over 300 miles an hour. If you monkey with the vapor pressure or the huge latent heat of vaporization, you get either too much cloud or none at all; either way, it's disaster. Try reducing the high specific heat of water, and remember there are few other substances that are that high, and you lose the stabilizing effect of the ocean surface and get more storms, colder average days. Virtually everything he tried either had very little effect, as with viscosity and conductivity, or else drastic effects in either direction."

"Then they were true optimums, Tom?" asked Dr. Takoa.

"Both ways," answered Feeney. "Whether you tilt the Earth up straight or lean it over to roll around like Uranus, you need another kind of water. Up straight, our water gives too much thermal stability and you get complete cloud cover and ice caps down to latitude 40 north and south. With the polar axis down flat, you get wild storms because of the source-sink effect of the hot and cold poles. To moderate this, you need a lower freezing point, lower specific heat so the storms are more localized, and other changes. In fact, it was the angle of tilt of the Earth's axis that optimized most sharply with the properties of water."

Commander Brunel sighed. "And what was the optimizing parameter, Tom?"

Father Feeney shrugged. "It was some complicated thing that Fullerton had worked out. He went to the biologists and your gang" —Feeney pushed a finger at Professor Bateson—"to decide on a mean Earth temperature at which metabolic, hydrocarbon activity would best be served, then figured in a statistical number that expressed the severity of wind and climatic variations, kind of a standard deviation for lousy weather."

"It was geared to us, Tom, to human life?" asked Brunel, frowning.

Feeney nodded. "Yeah . . . us, Dave . . . not snakes or ants. Things that run at 98.6 Fahrenheit."

But Commander Brunel was already wondering what he could possibly tell Houston in the next transmission.

Father Feeney continued. "What happened next, I have no excuses for. I don't know how much you remember about the university situation fifteen years ago, but it was a discouraging, demoralizing time. I know I wasn't thinking clearly. I hated the school, but I didn't know what else to do, or to put it in the language inside my head, what God *wanted* me to do.

"Fullerton had already done three dissertations' worth of work and neither he nor I was satisfied with what he had. Oh, it would have been another story three or four hundred years ago. They would have made him a Prince of the Church. How Rome would have delighted in all his simulation output as a counter to nasty old Copernicus. 'Look,' they would have said, 'so what if we aren't the center of the system? We're even more unique and unusual, and thus more God-centered.' Elias and I didn't see it like that. Planetary systems form in statistically allowed ways, as you well know, Dave. How freaky were we? That was the question Elias wanted to look at next. He wanted to go on and make a statistical study to determine the probability of planets that fell within the several limits apparently expressed by the properties of water. That is, planets of our mass, solar constant, tilt, primary star type . . . Yes, he had even looked at the sun's spectral class. You see, natural ocean water has a kind of spectral window to the most intense solar spectral region. . . . Oh, rotation speed was another critical parameter, and related to specific heat and latent heat of evaporation."

Commander Brunel smiled thinly. "It came out a damn small chance, didn't it, Tom?"

Father Feeney nodded. "The chance of a water-optimized Earth occurring in our own galaxy varied between one in ten-to-the-seventh to one in ten-to-the-ninth, depending on how far you were willing to let Elias's optimum 'life parameter' slip. In the visible universe, it was higher, a chance of one in ten to one in a thousand that a single Earth might appear."

He looked around at the others, his face heavy with remembered distress. "To understand what happened next, I have to tell you a little about what was going on then in the university. Oh, Notre Dame was nothing special; half the big schools and most of the private ones were on strike that year in one way or another. It isn't very priestlike to say it, but I'd really had a gut-full of that faculty. It was obvious that the country faced serious difficulties, changes in jobs, retrenchment, but all they wanted was more money and fewer hours to do the same, tired old stuff. The most despicable thing about them was the fact that they all knew their own students could never aspire to the kind of leisure and affluence they had lived through themselves, yet they were prepared to withhold grades, teaching and degrees to continue their own pleasant and unhurried lives.

"At first the students fell in with the faculty militants . . . It wasn't that they were stupid, just young and bewildered by the canniest con men of all time, full bulls on the liberal arts faculty. But a graduation missed because no grades come in tends to concentrate one's mind, like a hanging, as Samuel Johnson once said.

"By the time Elias finished his last calculation, the students were in court after restraint and had begun to picket the faculty pickets. It was ugly enough, all right, in and out of school. Shouting, curses, old men who had once lectured on Chaucer and Aquinas and John Locke spitting venom and hate at young persons who responded with rocks, shouts, and open cans of paint. That morning was especially bad, a car afire at one campus entrance, local and state police between the various groups, tear gas, and filthy, abusive language everywhere."

Feeney looked around. "If that was what universities had become, it seemed to me that we were lost and society ended. Where was the source of renewal or sanity? Of course they didn't bother me much. Some of the students cheered as I went through the lines in my clerical collar while the faculty pickets turned their backs. Still, I got phone calls all the time . . . threats and hatred . . . shouts about dirty scab Catholic finks . . . all that kind of phony talk the faculty had learned from reading about the earlier days of labor war, when real men fought for real rights. I'm telling you this so that you'll understand what comes next.

"Elias came to my office at about nine that day. His face was white and he seemed very disturbed. I was disturbed too, and when he showed me his results, I went into a kind of funk. What he was saying was that we, we on Earth, were all there was anywhere . . . an improbable oddity. A dreadful, small, selfish, hurting people, beset by greed and violence, and still . . . the absolute lords of the known universe! It was really shocking, unbelievable. I spent the whole day going over his stuff, assumption by assumption. We found one mistake and it only made the probabilities a little higher, a little worse.

"By five that afternoon there was loud chanting from the campus entrances and sirens were all about us. Elias Fullerton stared at me and his bony face was a mask of white distress. 'Are we really God's creatures?' he asked finally.

"I nodded, feeling sick. 'We are imperfect, Elias,' I said.

"He pointed out toward the sounds of riot. 'Those men and women represent the culmination of three thousand years of humanistic scholarship and philosophy. And all they can think about is greed and grossness . . . greed above all!'

"'Elias,' I tried to say, 'there are good men yet. They are silent now, but they are all around us.' But I didn't really believe it and he could see that. The mistake we were both making was in assuming that if people with PhD's could turn into bums, the rest of the world was already gone." Father Feeney chuckled. "As though Saint Francis or the folks who faced the lions had PhD's. Not likely!

"There really were good, steady citizens, millions of them, and in the end they muddled the country through to where we are now. That fuss at the universities was just a kind of specialized decadence, local, and the old bulls were really more to be pitied than hated.

"But back then everything seemed ended and awful. I have no excuse for what happened next. I should have seen that Elias was in deep trouble, but I was like those people out on the picket line, thinking of my own philosophical problems, my own alienation. I never really knew him. We should have had a drink together or gone to a movie sometime. He walked out into that wild night, white-faced, his hands shaking in rage and dismay, and all around us were sirens, violence, the flicker of fires across the campus.

"I decided to sleep in the department office that night. We had some cots set up in the basement corridor. Why didn't I keep Elias with me? I still think about that now and then. I certainly wasn't much of a priest. There were plenty of devils whispering in your ear at the university then!

"The next morning I was in my office early, thinking about what Elias had found and trying to make some other interpretation of it. There was still noise of riot, but old Father Ryan, the Dominican head of the department, walked in, sat down, and stared at me. 'Father Fullerton killed himself last night, Tom,' he said evenly. 'He went off a bridge into the river. He tied two flatirons to his ankles.'

"He had chosen death by water, the same way Quentin Compson killed himself in Faulkner's story. Elias liked Faulkner; it was about all he read outside of geophysics. I couldn't answer Father Ryan and the old man went on. 'He cleaned out his office, Tom, all his dissertation stuff. Apparently burned it all . . . plenty of fires last night for that, I guess.'

"I began to shake; my hands, my head. I suppose you three have never had that happen to you? If NASA had ever heard of it, I wouldn't be here now. My head seemed huge, expanded. My jaw muscles were absolutely locked, rigid, and I couldn't control the shaking. I began to cry and tried to speak. Finally I said, 'Father . . . Father . . . I can't go through those lines . . . I'll kill the bastards . . . Kill . . . I hate them . . . Father. . . .'

"'That old man was stooped and wrinkled, but he was tough. He got up and came behind my desk and took my arm and actually stood me up, like unfolding a jackknife.

"'I'll get you off the campus, Tom,' he said to me in a low voice. We walked to the campus police offices. I don't know what Ryan said to the head of Campus Security but within minutes we were in a police car, driven by a youngster who wore a gun, the red lights spinning on top, the siren screaming. We went through the faculty pickets at sixty miles an hour, like a cannonball, and I only caught a glimpse of shaken fists. A rock bounced off the trunk.

"We drove six hours that day, Father Ryan, the young cop, and I. In another state they left me at a Dominican retreat. They didn't give me food or sleep when I got out, still shaking, trying to weep from eyes long run dry. They took me into the chapel and I fell onto my knees with them and we were there a very long time.

"They got me through the worst of it, somehow, and a week later a young Jesuit in a rented car arrived and in two days I was at an exclusive retreat high in the mountains of New Mexico. They valued me, the Society, and they weren't going to flush their efforts and investments down the drain without a fight. I never went back to the university. They bought me a place on an eclipse expedition and I did some good things there. Then came NASA and all the rest of this. I've never told you folks the story for obvious reasons, though I guess by now I probably won't start shaking again."

Dr. Takoa put her small hand on Father Feeney's arm. "Dear Tom," she said, "how too bad that you and Father Fullerton could not see then this other interpretation of his finding." She shook her head. "Yet, how could you? How could anyone?"

Commander Brunel frowned at the small exobiologist. "What 'other interpretation,' Anna?" he asked darkly.

She winked at him. "Why, the obvious one, Dave. Water evidently has variable properties. The reason Father Fullerton found an exact optimization is not that the Earth is 'lucky' or 'statistically unusual.' The Earth's water was specifically designed for it. Titan's water, if Tom's predictions mean anything, was designed for it as well."

Brunel stared at the floor of the lander and grunted. He looked with a pleading expression at Professor Bateson. "Greg, for heaven's sake. How can water be different here from on Earth? I mean . . ."

Bateson shrugged. "The thing is, Dave, why not? This Titan water is actually closer to the way you would expect our water to behave if it followed similar compounds like hydrogen sulfide. It still has too wide a liquid range, but at least it boils lower on the temperature scale. Theoretically, water should boil at about minus 100 Celsius." Brunel started shaking his head but the chemist went on. "You see, all the theories of boiling, freezing, vapor pressure, whatever, depend on measurements. I mean, we know that water boils when the energy of molecular agitation exceeds the attractive forces between the liquid water molecules. So we calculate those bonding forces from kinetic theory. But if Earth water had been like this stuff, we certainly wouldn't have found it any more novel than what we have now. We'd just compute constants for the binding forces and go along with them. The point I'm making, Dave, is that there's nobody who can prove from, *say*, the laws of thermodynamics, that water *must* behave the way it does for us. Its behavior is a given, an experimental starting point."

The chemist rubbed his chin thoughtfully. "Let me ask you something, Dave. Did they ever test the properties of Mars water, or bring any back?"

Commander Brunel tried to remember all the experiments. He shook his head. "The robot diggers really never found any in the temperate zone. They tried to incubate any encysted life in the soil with water they brought along."

Anna Takoa chuckled to herself. "The poor Martian bacteria, Dave! Oh, dear! I suppose surface tension, capillarity, and osmotic pressures must be very different between these different waters. If there was any life in the Martian dirt samples, we probably either exploded it—" she suddenly puffed out her cheeks—"or shrank it away to nothing. . . ." And now she sucked in until her lips poked out like a guppy's. "No, Greg. I don't want *any* of that ice in my bourbon!"

"Wasn't there a Martian polar cap probe, Dave?" asked the chemist.

"Yes, but it was mainly for seismic and weather measurements.

Think about it, Greg. Why use up room and power on an expensive space craft to run high school experiments on water?"

"Which they wouldn't have believed from that distance anyway," continued Bateson in a thoughtful voice.

"Look. Do any of you have an explanation for this?" said Brunel in exasperation. "Are you willing to just accept that the damn stuff changes at will from planet to planet?"

Professor Bateson shrugged again, but he spoke quietly and seriously. "Dave, what else have we got but Tom's story? If this isn't a theological matter, then what else is it?"

Feeney's large, ruddy face was beaming at them now. "Think of it!" he said excitedly. "We're just talking about water. But what about carbon? Or silicon? Their properties aren't even completely known yet. Imagine what other world-to-world adjustments may also exist!"

Brunel shook his head fiercely. "Now wait a minute, all of you! Let's back off from the metaphysics and get back to physics. Greg, what if we disassociate the Titan water and burn its O_2 and hydrogen with our hydrogen and O_2? Do we get some kind of intermediate or average property?"

They all thought about that. Finally, Father Feeney frowned and rubbed his hands together. "Dave, I don't know. Do we want to try that right now . . . ?" His voice trailed off as Brunel peered at him in puzzlement.

Anna Takoa patted Tom Feeney's hand and cocked her head. "Tom, I think Dave has proposed a crucial experiment. We *must* go forward with this."

Feeney suddenly grinned at them and leaned back on his stool. "Of course we must. We've gotten this far following what we think is the path to truth. Now, if ever, we should go ahead," and his smile became radiant. "But, Greg, I do suggest that we test the decomposed and recombined Titan water first, to make sure we're not doing something by breaking down the molecule."

Professor Bateson clapped his hands in mock applause. "You know, I really do like being on a geriatric crew. Young kids never think of things like that. They just want to go ahead as fast and spectacularly as possible."

They formed a tense line in front of the hood, each on a stool, while Professor Bateson added catch tubes to his electrolysis apparatus, then led the gas flows to a combustion chamber and condensing apparatus that allowed the recombined water to drip into an analysis-box specimen-holder. Bateson turned up the juice on the electrolysis, and lit off the combining gas so that a blue flame flickered steadily and water droplets condensed and ran down the cooled glass of the condenser. Minutes passed and finally the sample holder was full. Bateson took a deep breath and shifted the water specimen to the analysis box. "Okay, here's good old density-versus-temperature again." The monitor screen flashed up a curve.

Commander Brunel exhaled and blinked. "Now . . . it's just water, our water." Indeed, the density curve lay exactly over the Earth-water reference, starting at zero C instead of minus 67 as before.

But Tom Feeney's eyes were wide and bright. "Greg . . . Greg . . ." and his voice was so tense that they all turned and stared at him. "Test some of the water from the beaker that didn't go over."

Bateson stared. "We've done the distilled Titan water three times already, Tom."

"Again, Greg. Please!"

Bateson turned without a word, put his hands in the gloves, and drew a sample from the original flask of distilled Titan water. He placed it in the analysis box and they silently watched as the data flashed up on the screen. Now it too was identical with the previous sample and the Earth reference curve.

Commander Brunel's cheeks turned dead white. "Now . . . hang on here! We weren't hallucinating! Those other experiments are all on video tape. That camera goes on every time Greg shoves his hands into the gloves!"

Tom Feeney stared at them in wonder. "Don't you see it yet? Any of you? He . . . He *designs* each world . . . to be as *gentle* as He can make it! It didn't work on Titan, maybe not on Mars either. Nothing came to live. But now *we're* here! And we're searching! *We're* the life on Titan! He's *given* Titan, its water, everything, to us! Think of it, Dave! He's like Johnnie Appleseed, preparing orchards all across the universe. And then changing them when a different seed comes by. How beautiful! Oh, Anna, think of the variety, the endless kinds of life! Ah . . . the love . . ." And Tom Feeney dropped to his knees on the floor of the lander and lifted his hands in prayer. He remembered a summer moment when he was fourteen as though it were an hour ago. He had gone with some Protestant friends to a Presbyterian song service on a Sunday night in the summer resort town where his family spent two weeks each year. They had a new minister, a bearded young man, who had organized a musical program with a choir and some soloists. The part of it that Tom Feeney remembered was a thin, plain girl who stood and sang in a sweet, round, innocent voice a song called "The Saints of God." Tom Feeney did not remember most of the verses; they were about different saints. . . . "One was a doctor, and one was a thief, and one was devoured by a fierce, wild beast. . . ." But he had never forgotten the refrain. . . . "And there's no real reason, no none in the least, why I shouldn't be one too," sung in that pure, lovely voice. Kneeling on the floor of the Titan lander, Father Thomas Feeney, S.J., suddenly, finally, understood how you became a saint. You *knew!* It was that simple. *You knew!* And the intensity of his prayers bloomed like sudden flowers.

But the rest of them were not yet ready. Anna Takoa peered intently at the two other men. "Send out the land runner! See if the ice sheet has changed too! Quickly!"

They darted over to the control station for the robot vehicle and soon they were watching a TV view of the runner scooting out on the ice sheet. They braked it over a smooth spot they had surveyed before and the digger descended and pulled a plug of ice up into the runner's innards. Brunel's hands moved over the console and they watched a density curve move upward on the screen. Earth water again!

The little woman's eyes were wide. "Have it put down the acoustic probe, Dave! That changed water must be freezing on the bottom right now!"

The acoustic profiler immediately extended and tracked the thickening ice sheet. There was no doubt about it. The water was rapidly freezing solid in the hot-spot bottom pools. The ice cover was thickening while they watched, at a rate of half a meter per minute. The bolometer monitors they had planted in deep bore holes in the ice were registering sudden, intense thermal output from the latent heat released by the solidifying ice.

Commander Brunel shut his eyes and let his head fall back to the steel wall behind him. "Greg . . . Anna . . . Tom. Good God! While we were testing a half liter of its water, something has changed the whole planet!"

But Anna Takoa had finally seen and heard enough to understand it. She fell to her knees in front of Tom Feeney and pressed her thin hands up between his. Her eyes filled with tears of joy. She said to him in her soft voice, "The Lord Buddha can appear as a running, tinkling brook as well as a man, Tom."

Feeling her hands between his, Father Feeney bent toward her small face. "Monet, the painter, knew that God lived in the pond and the lilies. Oh, think of a billion different ponds and lily pads, dear, dear Anna!"

"And frogs on them, Tom. Fat, green, croaking. All different. All beautiful!"

Their souls swooned together then and their prayers pierced the metal of the lander and flew outward into the great spangle of stars that lay in every direction.

And at just about that same time, hundreds or maybe thousands of other, more or less similar, groups were making these same discoveries and responding in nearly identical ways. So that for those who could see and delight in it, the entire universe was constantly pierced by bright new arrows of surprise and love and joy.